The Spectral Transformation Lanczos Algorithm for the Symmetric-Definite Generalized Eigenvalue Problem: A Comparative Analysis with Conditioning Insights

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Problem Statement

The Symmetric Definite Generalized Eigenvalue Problem

For $n \times n$ real matrices $A = A^{\mathrm{T}}$ and positive definite (or semidefinite) $B = B^{\mathrm{T}}$, find $\boldsymbol{v} \neq \mathbf{0}$ and λ such that

$$A\boldsymbol{v} = \lambda B\boldsymbol{v}$$

where $\mathcal{N}(A) \cap \mathcal{N}(B) = \{\mathbf{0}\}$. The value λ is a generalized eigenvalue and $\boldsymbol{v}$ is the corresponding generalized eigenvector. If B is invertible, the generalized eigenvalues are eigenvalues for $B^{-1}A\boldsymbol{v} = \lambda\boldsymbol{v}$. Otherwise, for $B\boldsymbol{v} = \mathbf{0}$, we say that $\lambda = \infty$ is an eigenvalue. If B is positive definite, there are n linearly independent eigenvectors.

We assume throughout that $A \neq 0$ and $B \neq 0$.

Applications and Algorithms I

- Generalized eigenvalues are real for this problem.
- Vibration Analysis in Structural engineering e.g (Boeing). For example, equation of a vibrating system:

$$K\mathbf{x} = \lambda M\mathbf{x}$$

  where M is the mass matrix, K is the stiffness matrix, $\mathbf{x}$ is the displacement vector, and λ is the eigenfrequencies of the system.
- Existing algorithms for both dense and sparse problems typically reduce to a symmetric standard eigenvalue problem.
- Reducing to a symmetric eigenvalue problem preserves symmetry and we get real eigenvalues.
- Stability has been a long-standing concern.

Applications and Algorithms II

- Recent work on direct methods have proven residual bounds for dense problems. [Michael Stewart, 2024].
- This work is about applying an iterative method to the sparse problem, and verifying the residual bounds predicted for direct methods.
- We start with some general background.

The Standard Reduction (J.H. Wilkinson, 1965)

If B is positive definite then it has a Cholesky factor

$$B = C_b C_b^{\mathrm{T}}.$$

Thus

$$A\boldsymbol{v} = \lambda B\boldsymbol{v} \Leftrightarrow A\boldsymbol{v} = \lambda C_b C_b^{\mathrm{T}}\boldsymbol{v} \Leftrightarrow C_b^{-1} A C_b^{-\mathrm{T}} C_b^{\mathrm{T}}\boldsymbol{v} = \lambda C_b^{\mathrm{T}}\boldsymbol{v}.$$

Standard Reduction to an Ordinary Eigenvalue Problem

Solve the symmetric eigenvalue problem

$$C_b^{-1} A C_b^{-\mathrm{T}}\boldsymbol{u} = \lambda\boldsymbol{u}, \qquad \boldsymbol{u} \neq \mathbf{0}$$

Then solve $C_b^{\mathrm{T}}\boldsymbol{v} = \boldsymbol{u}$.

This is a symmetric eigenvalue problem that can be solved using either a direct method or the Lanczos algorithm.

The Standard Reduction (cont'd)

- This shows that the symmetric definite generalized eigenvalue problem has real eigenvalues. Eigenvectors are orthogonal with respect to the inner product $(\boldsymbol{x}, \boldsymbol{y}) = \boldsymbol{y}^{\mathrm{T}} B \boldsymbol{x}$.
- For dense problems it is the approach used by LAPACK. It is also used frequently, but not universally, with the Lanczos algorithm for sparse problems.
- For efficiency on sparse problems, a sparse Cholesky factorization is required to use it with Lanczos.
- It fails if B is semidefinite and is unstable when B is ill-conditioned (i.e. $\kappa_2(B) = \|B\|_2 \|B^{-1}\|_2$ is large.)
- If B is ill-conditioned, it usually delivers small residuals for large eigenvalues and large residuals for small eigenvalues.

Other Algorithms

- QZ Algorithm, (C. B. Moler and G. W. Stewart, 1972) - Useful for non-symmetric dense problems. So it is not suitable for sparse symmetric problems since it does not preserve symmetry. It is stable but also relatively~~very~~ slow.
- Simultaneous Diagonalization of A and B, (S. Chandrasekaran 2000) - Uses simultaneous~~simulatenous~~ diagonalization of A and B using congruence to compute eigenvalues. It is slow and of uncertain stability~~expensive.~~

An Alternate Formulation

The generalized eigenvalue problem can be formulated in another way as follows:

The Generalized Eigenvalue Problem Version II

The eigenvalue problem can be written

$$\beta A\boldsymbol{v} = \alpha B\boldsymbol{v}$$

where $\boldsymbol{v} \neq \mathbf{0}$ and β and α are not both zero. The original formulation eigenvalues are given by $\lambda = \alpha/\beta$. Each eigenvalue is a nonunique pair (α, β) that can be scaled by $c \neq 0$. It can be identified with a subspace of $\mathbb{R}^2$ (or $\mathbb{C}^2$):

$$\mathcal{E} = \{c \cdot (\alpha, \beta) : c \in \mathbb{R}\}$$

Lanczos Algorithm I

- Iterative method for finding eigenvalues and eigenvectors of symmetric matrices that works by transforming the matrix into a smaller tridiagonal matrix that retains the extremal spectral properties of the original matrix
- Builds an orthonormal basis of the Krylov subspace associated with the matrix.
- The basis is constructed iteratively, with vectors $\mathbf{q}_1, \mathbf{q}_2, \ldots \mathbf{q}_n$ formed during n iterations, and stacked together to form an orthogonal matrix: $Q_n = [\mathbf{q}_1 | \mathbf{q}_2 | \ldots | \mathbf{q}_n]$
- For a symmetric matrix $A \in \mathbb{R}^{m \times m}$ and an initial vector $\mathbf{v}_1$, the algorithm constructs the n-dimnensional Krylov subspace

$$\mathcal{K}_n(A, \mathbf{v}_1) = \text{span}(\{\mathbf{v}_1, A\mathbf{v}_1, A^2\mathbf{v}_1, \ldots, A^{n-1}\mathbf{v}_1\}).$$

Lanczos Algorithm II

- The resulting orthonormal basis is used to form a tridiagonal matrix

$$T_n = \begin{pmatrix} \gamma_1 & \delta_1 & & & \\ \delta_1 & \gamma_2 & \delta_2 & & \\ & \delta_2 & \gamma_3 & \delta_3 & \\ & & \ddots & \ddots & \vdots \\ & & & \delta_{n-1} & \gamma_n \end{pmatrix}$$

  whose eigenvalues approximate those of A.

- In general, the Lanczos algorithm computes a decomposition

$$AQ_n = Q_n T_n + \delta_n \mathbf{q_{n+1}} \mathbf{e_n}^T$$

Overview of Spectral Transformation I

- Spectral Transformation is a technique that is used to modify the spectrum of a matrix in a controlled way.
- It is usually done to improve convergence for an algorithm or to make certain matrix properties more accessible.
- For a standard eigenvalue problem

$$A\mathbf{v} = \lambda\mathbf{v}, \qquad \mathbf{v} \neq \mathbf{0}$$

  It transforms the problem into a shifted and inverted version

$$(A - \sigma I)^{-1}\mathbf{v} = \frac{1}{\lambda - \sigma}\mathbf{v}, \qquad \mathbf{v} \neq \mathbf{0}$$

  where $\sigma \in \mathbb{R}$ is the shift.

Overview of Spectral Transformation II

- Shifting makes the eigenvalues near σ more prominent in the new spectrum and also has the advantage that it preserves eigenvectors.
- One can apply a direct or an iterative method to the shifted and inverted problem.
- Ideal when focusing on eigenvalues near a specified shift σ.
- For the generalized eigenvalue problem involving **general matrices**

  $$A\mathbf{v} = \lambda B\mathbf{v}, \qquad \mathbf{v} \neq \mathbf{0}$$

  The spectral transformed problem will be

  $$(A - \sigma B)^{-1} B\mathbf{v} = \frac{1}{\lambda - \sigma}\mathbf{v}, \qquad \mathbf{v} \neq \mathbf{0}$$

Spectral Transformation for the Symmetric-Definite Problem

- For a symmetric-definite generalized eigenvalue problem $(A - \sigma B)^{-1}B$ is not **symmetric**!
- Can we obtain a symmetric formulation with the spectral transformation? **YES!**
- Spectral Transformation when applied to a symmetric-definite generalized eigenvalue problem, can be designed to preserve symmetry by employing a symmetric decomposition of B.
- We explore this in detail.

Spectral Transformation Lanczos [T. Ericsson and A. Ruhe, 1980]

Lemma

Let $\lambda = \alpha/\beta \neq \infty$ *and* $\boldsymbol{v} \neq 0$ *satisfy* $A\boldsymbol{v} = \lambda B\boldsymbol{v}$. *Assume that* $A - \sigma B$ *is nonsingular and* $B = C_b C_b^{\mathrm{T}}$, $C_b \in \mathbb{R}^{n\times r}$ *with linearly independent columns. Then* $\theta = 1/(\lambda - \sigma)$ *is an eigenvalue of the shifted and inverted problem*

$$C_b^{\mathrm{T}}(A - \sigma B)^{-1} C_b \boldsymbol{u} = \theta \boldsymbol{u}, \qquad \boldsymbol{u} \neq \boldsymbol{0}.$$

with eigenvector $\boldsymbol{u} = C_b^{\mathrm{T}}\boldsymbol{v} \neq \boldsymbol{0}$.
Conversely, assume that $\boldsymbol{u} \neq \boldsymbol{0}$ *is an eigenvector for the shifted and inverted problem with eigenvalue* θ. *Then the vector* $\boldsymbol{v} = (A - \sigma B)^{-1} C_b \boldsymbol{u} \neq 0$ *is an eigenvector for the eigenvalue* $(1 + \sigma\theta, \theta)$ *of the original problem.*

Spectral Transformation for Dense Problems [Michael Stewart, 2024]

This direct method employs the spectral transformation described by [T. Ericsson and A. Ruhe, 1980], and symmetric decompositions of $A - \sigma B$ and B such that

$$A - \sigma B = C_a D C_a^{\mathrm{T}}, \qquad \text{and} \qquad B = C_b C_b^{\mathrm{T}},$$

to transform the problem into a symmetric standard eigenvalue problem given by

$$C_b^{\mathrm{T}} C_a^{-T} D_a C_a^{-1} C_b \boldsymbol{u} = \theta \boldsymbol{u}, \qquad \boldsymbol{v} = C_a^{-\mathrm{T}} D_a C_a^{-1} C_b \boldsymbol{u}$$

with $(\alpha, \beta) = (1 + \sigma\theta, \theta)$ or $\lambda = (1 + \sigma\theta)/\theta$.

- B can factored using pivoted Cholesky and A using LDL^{T} factorization with rook pivoting, both available in LAPACK.
- We cannot expect a shift to result in well conditioned $A - \sigma B$ or C_a, **but ill conditioning is not what matters!**

Interesting Questions

- Residual bounds were proven for the direct method by [Michael Stewart, 2024], which employs spectral transformation.
- Do these residual bounds proven apply when an iterative method is used for the spectral transformed problem?
- How does symmetry in the decomposition of $A - \sigma B$ impact the spectral transformation Lanczos algorithm? ~~Does the spectral transformed problem respects symmetry in the decomposition of $A - \sigma B$?~~

What is our approach?

- Apply the Lanczos algorithm to the spectral problem
- Investigate if the residuals for the computed eigenvalues follows the bounds for the direct methods in terms of the choice of shift.
- Explore the effect of preserving symmetry in the decomposition of $A - \sigma B$ on residuals.

Spectral Transformation Lanczos Algorithm I

function $\textsc{Spectral\_Lanczos}(A, B, m, n, \sigma, tol)$

  Choose an arbitrary vector $\mathbf{b}$ and set an initial vector $\mathbf{q_1} = \mathbf{b}/\|\mathbf{b}\|_2$

  Set $\beta_0 = 0$ and $\mathbf{q_0} = \mathbf{0}$

  Set $Q = \text{zeros}(m, n+1)$

  Precompute the LU factorization of $A - \sigma B$: $LU = (A - \sigma B)$

  Factor: $B = CC^T$

  for $j = 1, 2, \ldots, n$ **do**

    $Q[:, j] = \mathbf{q}_j$

    $\mathbf{u} = C\mathbf{q}_j$

    Solve: $(LU)\mathbf{v} = \mathbf{u}$ for $\mathbf{v}$

Spectral Transformation Lanczos Algorithm II

$\mathbf{v} = \mathbf{v} - \beta_{j-1}\mathbf{q}_{j-1} - \alpha_j \mathbf{q}_j$

Full reorthogonalization: $\mathbf{v} = \mathbf{v} - \sum_{i \le j} (\mathbf{q}_i^T \mathbf{v})\mathbf{q}_i$

$\beta_j = \|\mathbf{v}\|_2$

if $\beta_j < tol$ **then**

restart or **exit**

end if

$\mathbf{q}_{j+1} := \mathbf{v}/\beta_j$

end if

end for

$Q = Q[:, 1 : n]$

$\mathbf{q} = Q[:, n]$

return $(Q, T, \mathbf{q})$

end function

Some Definitions from [Michael Stewart, 2024]

$$X = C_a^{-1} C_b, \qquad W = X^{\mathrm{T}} D_a X, \qquad \mu = \frac{\|X\|_2^2}{\|W\|_2} \geq 1.$$

$$\eta = \frac{\|A - \sigma B\|_2^{1/2}}{\|B\|_2^{1/2}}, \qquad \sigma_0 = \sigma \frac{\|B\|_2}{\|A\|_2}, \qquad \text{and} \qquad \gamma = \frac{\|A\|_2}{\|A - \sigma B\|_2}.$$

- The shifted and inverted problem is $W\boldsymbol{u} = \theta \boldsymbol{u}$.
- The only "inversion" is in solving $C_a X = C_b$.
- The values of μ, $\eta\|X\|_2$, σ_0, and γ can potentially impact stability.
- $\eta\|X\|_2$ is the most interesting and important of these.

Bounds for $\eta\|X\|_2$

We have

$$\eta^2\|X\|_2^2 \le \mu\kappa_2(A-\sigma B)$$

and even better

Lemma

Assume that $\sigma \neq 0$ and $A-\sigma B$ is invertible. Then

$$\begin{aligned}\eta^2\|X\|_2^2 &\le \left(1+\frac{1}{|\sigma_0|}\right)\frac{\mu}{\min_i\left|1-\frac{\lambda_i}{\sigma}\right|}\\ &= (1+|\sigma_0|)\frac{\mu}{\min_i\left|\frac{\|B\|_2}{\|A\|_2}\lambda_i-\sigma_0\right|}.\end{aligned}$$

Forward Errors

- **The size of $\eta^2\|X\|_2^2$ determines stability and it is usually much smaller than $\kappa_2(A-\sigma B)$.**
- It is surprisingly easy to avoid large $\eta\|X\|_2$. In practice, if $|\sigma_0|$ is not small, $\eta\|X\|_2$ is large only if σ is chosen to match an eigenvalue λ to several digits. A random shift in a reasonable interval almost always works.
- If A and B are both positive definite, all generalized eigenvalues are positive, $\mu = 1$, and simply choosing $\sigma_0 = -1$ gives

$$\eta^2\|X\|_2^2 \leq \left(1 + \frac{1}{1}\right) \frac{1}{\min_i \left|1 + \frac{|\lambda_i|}{|\sigma|}\right|} \leq 2$$

  This is a very common special case in structural engineering.

Error Bounds: Moderate Shifts and Eigenvalue Stability

Eigenvalue Backward Errors

For the computed θ_i, there exist symmetric E and F and a vector $\tilde{\boldsymbol{v}}_i \neq \mathbf{0}$ such that

$$\theta_i(A+E)\tilde{\boldsymbol{v}}_i = (1+\sigma\theta_i)(B+F)\tilde{\boldsymbol{v}}_i$$

and

$$\max\left(\frac{\|E\|_2}{\|A\|_2}, \frac{\|F\|_2}{\|B\|_2}\right) \leq O(u)(1+|\sigma_0|)\eta^2\|X\|_2^2 + O(u^2)$$

If $|\sigma_0|$ is not large and $\eta^2\|X\|_2^2$ is not large, each $(1+\sigma\theta_i, \theta_i)$ is an eigenvalue of a pair close to (A, B).

Error Bounds: Moderate Shifts with Computed Eigenvectors

Computed Eigenvector Bounds

There exist symmetric E and F such that the computed θ_i and the computed eigenvector $\boldsymbol{v}_i$ satisfy

$$\theta_i(A+E)\boldsymbol{v}_i = (1+\sigma\theta_i)(B+F)\boldsymbol{v}_i$$

with

$$\max\left(\frac{\|E\|_2}{\|A\|_2}, \frac{\|F\|_2}{\|B\|_2}\right) \leq$$
$$O(u)|1-\lambda_i/\sigma||\sigma_0|\left(1+\max(\gamma,1)\left(1+|1-\lambda_i/\sigma|\right)\eta^2\|X\|_2^2\right)+O(u^2)$$

If $|\sigma_0|$ and $\eta^2\|X\|_2^2$ are not large, $A-\sigma B$ does not cancel, and $\lambda_i = (1+\sigma\theta_i)/\theta_i$ is not much larger than σ, then each $(1+\sigma\theta_i, \theta_i)$ and $\boldsymbol{v}_i$ is an eigenvalue/eigenvector of a pair close to (A, B).

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with

$$\max\left(\frac{\|E\|_2}{\|A\|_2}, \frac{\|F\|_2}{\|B\|_2}\right) \leq$$
$$O(u)|1-\sigma/\lambda_i|\left(1+\max(\gamma,1)\left(1+|1-\lambda_i/\sigma|\right)\eta^2\|X\|_2^2\right)+O(u^2)$$

If $\eta^2\|X\|_2^2$ is not large, $A-\sigma B$ does not cancel, and $\lambda_i = (1+\sigma\theta_i)/\theta_i$ is not much larger or smaller than σ, then each $(1+\sigma\theta_i, \theta_i)$ and $\boldsymbol{v}_i$ is an eigenvalue/eigenvector of a pair close to (A,B).

Do these residual bounds applies to the ST Lanczos Algorithm?

- We have previously reviewed the residual bounds for a direct method.
- *To what extent do these bounds carry over to the Spectral Transform Lanczos algorithm?*
- Understanding how residual behaviour translates helps validate ST Lanczos in practice.
- Guides us in accessing convergence and numerical stability.

Setting up a Generalized Eigenvalue Problem

Given a diagonal matrix $D \in \mathbb{R}^{m \times m}$ with predefined eigenvalues, and a regularization hyperparameter δ, the following algorithm sets up a generalized eigenvalue problem

function $\textsc{Generate\_Matrix}(D, \delta)$
    Set $m = \texttt{size}(D)$
    Q, _ = qr(random.randn(m, m))
    $C = QDQ^T$
    $L_0 = \texttt{tril}(\texttt{random.randn}(m,\ m))$
    $B = (L_0 L_0^T) + \delta I$
    $L = \texttt{cholesky}(B)$
    $A = LCL^T$
    return $(A,\ B)$
end function

Numerical Experiments I

- Generate a diagonal matrix $D \in \mathbb{R}^{3000\times 3000}$ of eigenvalues using a uniform distribution in the range $(10^{-3}, 10^{7})$
- D is composed of 3 groups of eigenvalues with the following subranges
  1. Group 1 contains 1000 eigenvalues in the range $(10^{-3}, 10^{1})$
  2. Group 2 contains 1000 eigenvalues in the range $(10^{1}, 10^{3})$
  3. Group 3 contains 1000 eigenvalues in the range $(10^{3}, 10^{7})$
- Set regularization hyperparameter $\delta = 10^{-3}$(*ill-conditioned*).
- Generate matrices A and B with `GENERATE_MATRIX` function so that

$$\kappa_2(A) = 4.25 \times 10^{15}, \qquad \|A\|_2 = 1.17 \times 10^{11}$$
$$\kappa_2(B) = 8.08 \times 10^{6}, \qquad \|B\|_2 = 1.34 \times 10^{5}$$

Numerical Experiments II

- Both matrices A and B are positive definite with their eigenvalues $\Lambda(A, B)$ equal to the diagonal elements of D.
- We run Lanczos algorithm for $n = 2000$ iterations.
- We use a tolerance of 10^{-9} for the Ritz pair convergence for all experiments.

Relative Residuals I

- Relative Decomposition Residual

$$\frac{\|C_b^T(A-\sigma B)^{-1}C_bQ_n - Q_nT_n - \delta_n\mathbf{q}_{n+1}\mathbf{e}_n^T\|}{\|C_b^T(A-\sigma B)^{-1}C_b\|}$$

- Generalized Relative Residuals

$$\|\tilde{\mathbf{r}}_i\| = \frac{\|(\beta_iA-\alpha_iB)\mathbf{v}_i\|}{(|\beta_i|\|A\|+|\alpha_i|\|B\|)\|\mathbf{v}_i\|}$$

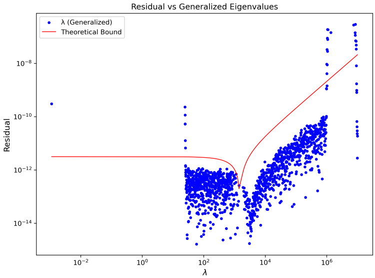
- Spectral Transform Residuals

$$\frac{\|C_b^T(A-\sigma B)^{-1}C_b\mathbf{u}_i-\theta_i\mathbf{u}_i\|}{(\|C_b^T(A-\sigma B)^{-1}C_b\|+|\theta_i|)\|\mathbf{u}_i\|}$$

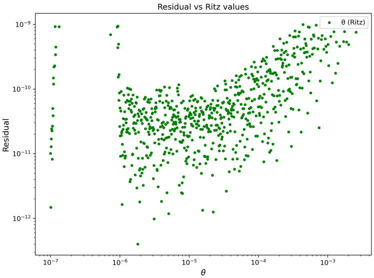
- Best Residuals

$$\frac{\|C_b^T(A-\sigma B)^{-1}C_b\mathbf{u}_i-\theta_i\mathbf{u}_i\|}{(\|C_b^T(A-\sigma B)^{-1}C_b\|+|\theta_i|)\|\mathbf{u}_i\|}$$

LU Decomposition (Moderate Shift with ill-conditioning) I



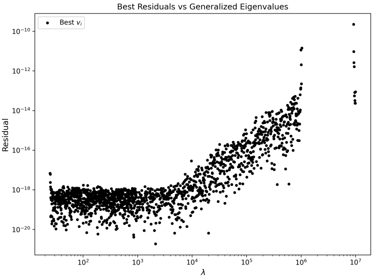
(a)



(b)

Figure: Residual plot with moderate shift $\sigma = 1.5 \times 10^3$ and $\kappa_2(A - \sigma B) = 5.5 \times 10^{12}$

LU Decomposition (Moderate Shift with ill-conditioning) II

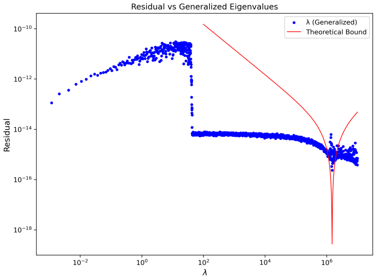


(c)

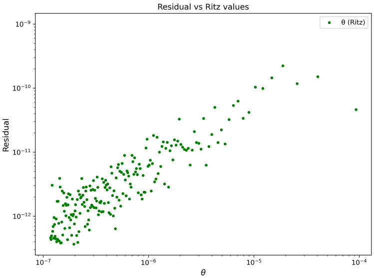
The computation gave a Lanczos decomposition residual of 1.30×10^{-7}. Plot (a) shows the generalized relative residual. The curve is approximation of the plot of $10^{-12}|1 - \lambda_i/\sigma|$. Plot (b) is the relative Ritz residuals. Plot (c) is the best achievable residual for an idealized eigenvector.

LU Decomposition (Large Shift with ill-conditioning) I

Figure: Residual plot with large shift $\sigma = 1.5 \times 10^6$ and $\kappa_2(A - \sigma B) = 3.10 \times 10^8$

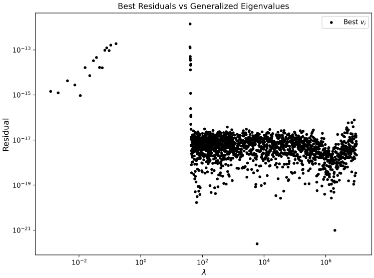


(a)



(b)

LU Decomposition (Large Shift with ill-conditioning) II

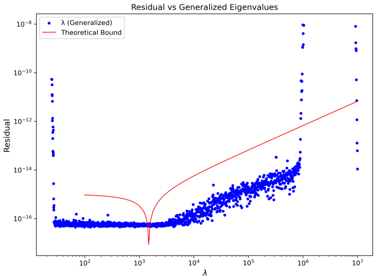


(c)

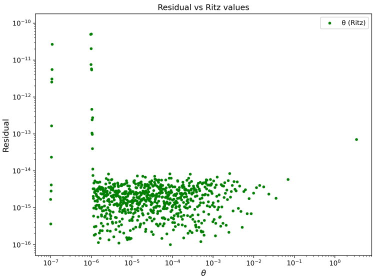
For a large shift, we obtain a Lanczos decomposition residual of as 6.14×10^{-10}. Plot (a) shows the generalized relative residual with the curve approximately given by $10^{-14}|(1 - \lambda_i/\sigma)(1 - \sigma/\lambda_i)|$. Plot (b) is the relative Ritz residuals. Plot (c) shows that eigenvalues that several orders of magnitude smaller than the shift cannot achieve a small residuals, except for those close to the shift.

Eigenvalue Decomposition (Moderate Shift with ill-conditioning) I

Figure: Residual plot with moderate shift $\sigma = 1.5 \times 10^3$ and $\kappa_2(A - \sigma B) = 5.5 \times 10^{12}$

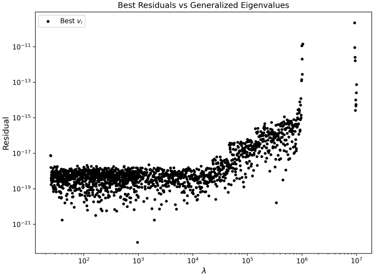


(a)



(b)

Eigenvalue Decomposition (Moderate Shift with ill-conditioning) II



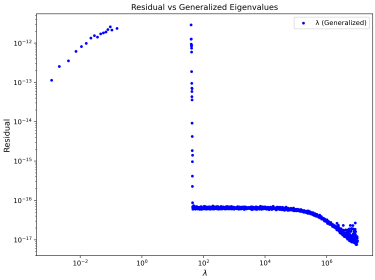
(c)

Eigenvalue Decomposition (Moderate Shift with ill-conditioning) III

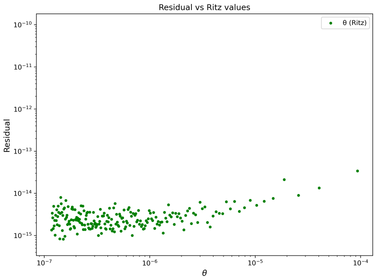
For a symmetric decomposition, we obtain a Lanczos decomposition residual of 10^{-27} which is significantly lower than that of a non-symmetric decomposition. Plot (a) shows the generalized relative residual. The curve is approximately a plot of $10^{-15}|1 - \lambda_i/\sigma|$. Eigenvalues not much lower than the shift achieves a residual of unit precision ($u \approx 10^{-16}$). Plot (b) shows the relative Ritz residuals. Plot (c) shows the best achievable residual for an idealized eigenvector.

Eigenvalue Decomposition (Large Shift with ill-conditioning I

Figure: Residual plot with large shift $\sigma = 1.5 \times 10^6$ and $\kappa_2(A - \sigma B) = 3.10 \times 10^8$

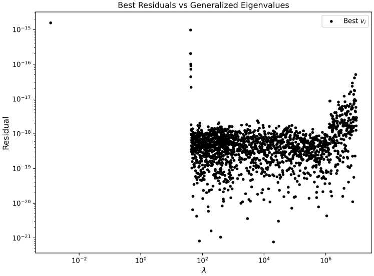


(a)



(b)

Eigenvalue Decomposition (Large Shift with ill-conditioning II



(c)

Eigenvalue Decomposition (Large Shift with ill-conditioning III

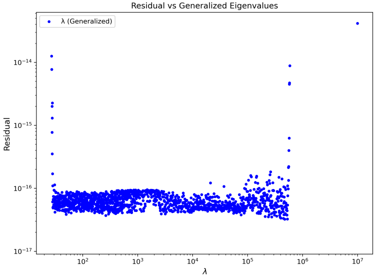
For a large shift, we obtain a Lanczos decomposition residual of order 10^{-32}. Plot (a) shows the generalized relative residual. Eigenvalues of orders of magnitude smaller than the shift achieves large residuals, with those close to eh shift achieving small residuals. Plot (b) shows the relative Ritz residuals. Plot (c) shows that eigenvalues that are several orders of magnitude smaller than the shift cannot achieve a large residual, except for those close to the shift.

Does the conditioning of $A - \sigma B$ affects stability?

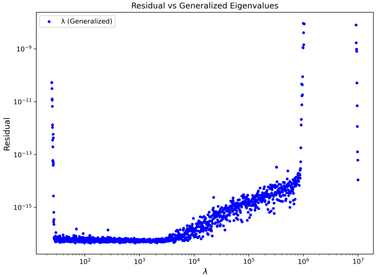
- Does conditioning affects stability for a symmetric decomposition and LU decomposition?
- Does conditioning affect~~affects~~ the residual bounds?
- Set $\delta = 10^{-12}$ to increase the conditioning of $A - \sigma B$ and compare generalized residuals for a moderate shift $\sigma = 1.5 \times 10^3$ with those obtained when $\delta = 10^{-3}$

Eigenvalue Decomposition —More Conditioning I

Figure: Residual plot comparison with a moderate shift $\sigma = 1.5 \times 10^3$



(a)

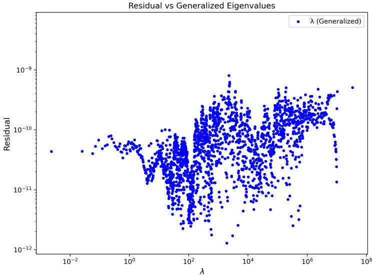


(b)

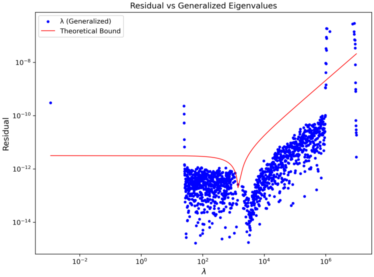
Eigenvalue Decomposition —More Conditioning II

Plot (a) shows the residual plot for $\kappa_2(A - \sigma B) = 3.75 \times 10^{18}$ while Plot (b) shows the residual plot for $\kappa_2(A - \sigma B) = 5.5 \times 10^{12}$. Eigenvalues not much smaller than σ delivers small residuals in both cases, which is as expected by the residual bounds, indicating that the symmetric decomposition is stable.

Figure: Residual plot comparison with a moderate shift $\sigma = 1.5 \times 10^3$



(a)



(b)

LU Decomposition —More Conditioning II

Plot (a) shows the residual plot for $\kappa_2(A - \sigma B) = 3.75 \times 10^{18}$ while Plot (b) shows the residual plot for $\kappa_2(A - \sigma B) = 5.5 \times 10^{12}$. It is observed that residual bounds were greatly affected by the conditioning of the system as seen from Plot (a). This implies that non-symmetric factorization of $A - \sigma B$ does not stablize the Lanczos process.

Summary

- Moderate shift σ that is not too close to a generalized eigenvalue will guarantee small residuals for eigenvalues not much larger than σ, but a large shift will result in having small residuals only for eigenvalues near σ. This is what we expect from the residual bounds for direct methods.
- Symmetric factorization stabilizes the ST-Lanczos algorithm.
- Stability of non-symmetric factorization depends on the conditioning of $A - \sigma B$

Pros and Cons

Pros:

- The algorithm is fast for sparse matrices since it uses Lanczos algorithm which is $O(nm^2 + n^2m)$.
- It is efficient in computing a subset of eigenvalues, making it memory efficient.
- With a little effort, it can be designed to handles the case of semidefinite B effectively.
- Delivers really small residuals for symmetric decompositions.

Cons:

- The generalized eigenvector computation is not unconditionally stable.

Future work

- Provide a mathematical justification for the improved stability of spectral transformation methods when a symmetric factorization of $A - \sigma B$ is used.

Thank you!